

Formula

$$\bar{x} \pm t_{\{\alpha/2, df\}} \cdot \frac{s}{\sqrt{n}}$$

where $df = n - 1$.

Example

Quiz sample: $\bar{x} = 81.6$, $s = 6.3$, $n = 12$, 95% CI uses

$$t_{\{0.025, 11\}} = 2.201 \Rightarrow 81.6 \pm 2.201 \cdot 6.3 / \sqrt{12} \Rightarrow (77.6, 85.6)$$

Interpretation

“We are 95% confident that the true mean score lies between

77.6 and 85.6.” Since 85 is within, the two-tailed test at $\alpha = 0.05$ fails to reject H_0 .